

TIME AND DISTANCE

Directions for questions 1 and 2: Select the correct alternative from the given choices.

A train travels 36 km in 15 minutes. Its speed is _____.

☐ a) 20 m/s

☐ b) 30 m/s

☒ c) 40 m/s (Your answer - ✓)

☐ d) 25 m/s

Solution:

$$\frac{36 \text{ km}}{15 \text{ min}} = \frac{36000}{15(60)} \frac{\text{m}}{\text{s}} = 40 \text{ m/s}$$

Choice (C)

If a man is travelling at 36 kmph, how many metres does he cover in 40 seconds?

☐ a) 400

☐ b) 300

☐ c) 2400

☐ d) 240

Solution:

$$36 \frac{\text{km}}{\text{hr}} = \frac{36,000 \text{ m}}{3,600 \text{ s}} = \frac{10 \text{ m}}{\text{s}}$$

In 40s, the distance covered is 400m.

Choice (A)

Directions for question 3: Type in your answer in the input box provided below the question.

The speed of a train is 15 m/s. The time it takes to cross a telephone pole is 10 seconds. What is the length of the train (in m)?

Your Answer: Unattempted

Correct Answer: 150 ✓

Directions for question 4: Select the correct alternative from the given choices.

A man covers the distance between A and B in 30 minutes and returns from B to A in 60 minutes. If the distance between A and B is 40 km, what is his average speed?

- ☐ a) 26 kmph
- ☐ b) $53\frac{1}{3}$ kmph
- ☐ c) $66\frac{2}{3}$ kmph
- ☐ d) 70 kmph

Solution:

The man covers 80 km in (30 + 60) min, the average speed for the round trip is $\frac{80(2) \text{ km}}{3 \text{ hr}} = 53\frac{1}{3} \frac{\text{km}}{\text{hr}}$. Choice (B)

Directions for question 5: Type in your answer in the input box provided in the question.

A certain distance is covered at a certain speed. If half of the same distance is covered in double the time, the ratio of the former speed to the latter is

Solution:

A distance, say $2d$, is covered in time, say t .
The speed is $2d/t$. Half the distance, i.e. d , is covered in time $2t$. The speed is $d/2t$.
The ratio of the speeds is $2 : \frac{1}{2} = 4 : 1$ Ans : (4)

Directions for questions 6 and 7: Select the correct alternative from the given choices.

A man can row downstream at 12 kmph and upstream at 8 kmph. Find the ratio of the speed of the current to the speed of the man in still water.

- ☐ a) 1 : 5
- ☐ b) 5 : 4
- ☐ c) 25 : 16
- ☐ d) 16 : 25

Solution:

If the man's speed in still water and the speed of the stream are v and u respectively, the downstream and upstream speeds are $v + u$ and $v - u$ respectively.
 $v + u = 12$ km/hr and $v - u = 8$ km/hr
 $\therefore v = 10$ km/hr, $u = 2$ km/hr and $\frac{u}{v} = \frac{2}{10} = \frac{1}{5}$ Choice (A)

Directions for questions 6 and 7: Select the correct alternative from the given choices.

Some telegraph poles are placed 20 m apart. How many poles will a train pass in three hours
at a speed of 60 kmph?

- ☐ a) 1200
- ☐ b) 2500
- ☐ c) 4000

☐ d) 9000 (Correct Answer - ✓)

Solution:

The speed of $60 \frac{\text{km}}{\text{hr}}$ is the same as $\frac{60(1000)}{3600} \frac{\text{m}}{\text{s}} = \frac{50}{3} \frac{\text{m}}{\text{s}} = 16\frac{2}{3} \frac{\text{m}}{\text{s}}$.

In 3 hours, i.e. $3(3600)$ s, the train would cover $\frac{50}{3}(3)(3600)$ m, i.e. 180,000 m.

There would be 9000 poles spanning this distance.

Choice (D)

Directions for questions 8 and 9: Type in your answer in the input box provided below the question.

An escalator moves from the ground floor towards the first floor at a rate of 10 ft/sec and its length is 120 feet. If a person walks on the moving escalator at a rate of two feet per second towards the first floor, how much time (in seconds) does he take to cover the entire length?

Solution:

The escalator moves up at a speed of 10 ft/sec. The person walks up, with respect to the escalator, at 2ft/sec. Therefore, his speed with respect to the ground is 12ft/sec. To cover the distance of 120 feet, he takes 10 seconds.

Ans : (10)

Directions for questions 8 and 9: Type in your answer in the input box provided below the question.

A, B and C run along a circular track having a length of 1.2 km, with speeds of 6 km/hr, 8 km/hr and 9 km/hr respectively. A and B run in the same direction but C runs in the opposite direction. If they all start at the same time and from the same place, how many times will A and C meet anywhere on the track by the time A and B meet for the first time anywhere on the track?

Solution:

A and B start from the same point, say P, on the circular track and run in the same direction, say clockwise, at 6 km/hr and 8 km/hr respectively. For the two to meet, B (the faster one) has to gain one entire round with respect to A. This would take $1.2 \text{ km} / 2 \text{ km/hr}$, i.e. 36 min.

C starts from P, at the same time as A and B and runs anticlockwise, at 9 km/hr.

A and C would meet in $1.2 \text{ km} / 15 \text{ km/hr}$, i.e. $\frac{12}{150}(60) \text{ min}$, i.e. 4.8 min.

Therefore A and C meet $\left[\frac{36}{4.8} \right]$, i.e. 7 times by the time A and B meet.

Ans : (7)

Q10

Directions for question 10: Type in your answer in the input box provided in the question.

In a 1000 metre race, A beats B by 50 m. In a 3 km race A beats B by m.

Solution:

In the time that A covers 1000 m, B covers 950m.

$\therefore$ In the time that A covers 3000 m, B could cover 2850 m, i.e. A would beat B by 150 m.

Ans : (150)

Q11



Directions for questions 11 to 13: Select the correct alternative from the given choices.

Two trains started at the same time from two stations towards each other. Their speeds are 72 kmph and 64 kmph respectively. By the time the two trains meet, one had covered 72 km more than the other. What is the distance between the two stations?

- ☐ a) 1208 km
- ☐ b) 1224 km
- ☐ c) 1280 km
- ☐ d) 1244 km

Solution:

Let the two trains be A and B. In one hour A covers 8 km more than B. To cover 72 km more than B, the two trains would have to travel for 9 hours. In these 9 hours, A and B together cover the distance between the stations. This distance has to be $(72 + 64) \div 9$ km, i.e. $(136) \div 9$ km, i.e. 1224 km.
Choice (B)

Directions for questions 11 to 13: Select the correct alternative from the given choices.

Raju takes six hours to walk to a certain place and ride back. He would have gained $2\frac{1}{2}$ hours, had he ridden both ways. How long would he take to walk both ways?

- ☐ a) $4\frac{1}{2}$ hours
- ☐ b) 6 hours
- ☐ c) $8\frac{1}{2}$ hours
- ☐ d) 7 hours

Solution:

Let the time to walk the distance and ride the distance be a hours and b hours respectively.
 $a + b = 6$
 $b + b = 3.5$
 $\therefore b = 1.75$ and $a = 4.25$.
To walk both ways, the time taken would be $2(4.25)$, i.e. 8.5 hours.
Choice (C)

Directions for questions 11 to 13: Select the correct alternative from the given choices.

In a 1000 metre race, if A gives B a head start of 100 metres, both of them reach the finish line

at the same time. If A's speed is 20 m/s, what is B's speed?

- ☐ a) 16 m/s
- ☐ b) 17 m/s
- ☐ c) 18 m/s
- ☐ d) 19 m/s

Solution:

A covers 1000 m in the time that B covers 900m
∴ A covers 20 m in the same time that B covers 18 m.
As A's speed is 20 m/s, B's speed has to be
18 m/s.

Choice (C)

Q14



Directions for question 14: Type in your answer in the input box provided below the question.

A man can row at 6 kmph in still water and a river is flowing at 4 kmph. How long (in minutes) will the man take to go to a place 1 km downstream and return?

Solution:

The man's speed in still water is 6 km/hr while the speed of the stream is 4 km/hr. To cover a distance of 1 km downstream and 1 km upstream, he would take $(\frac{1}{6+4} + \frac{1}{6-4})$ hr, i.e. $(\frac{1}{10} + \frac{1}{2})$ hr i.e. 36 min.

Ans : (36)

Q15



Directions for question 15: Select the correct alternative from the given choices.

The speeds of two persons are in the ratio 3 : 2. If the first person takes ten minutes less to cover a distance, what is the time taken by the second person to cover the same distance?

☐ a) 20 minutes

☒ b) 30 minutes (Your answer - ✓)

☐ c) 10 minutes

☐ d) 50 minutes

Solution:

The speeds of the 2 persons, say A and B are in the ratio 3 : 2. The time taken to cover a certain distance would be in the ratio 2 : 3. As A takes 10 min less than B, A takes 20 min while B takes 30 min.

Choice (B)